

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven questions only):**[2 x 7 = 14]**

- (a) An algorithm is said to be *finite* if
- (i) It uses finite memory
 - (ii) It produces correct output
 - (iii) It has finite input
 - (iv) It terminates after a finite number of steps
- (b) In Master's theorem, if $f(n) = \Theta(n^{\log_b a})$, then the solution of the recurrence is
- (i) $\Theta(f(n))$
 - (ii) $\Theta(f(n)\log n)$
 - (iii) $\Theta(n^{\log_b a})$
 - (iv) $\Theta(\log n)$
- (c) What is the maximum number of swaps that can be performed in the Selection Sort algorithm?
- (i) n-1
 - (ii) n
 - (iii) 1
 - (iv) None of the above
- (d) The time complexity of Strassen's Matrix Multiplication is approximately
- (i) $O(n^3)$
 - (ii) $O(n^2)$
 - (iii) $O(n^{2.81})$
 - (iv) $O(n \log n)$
- (e) Which of the following problems can *not* be solved optimally by a greedy approach?
- (i) Minimum Cost Spanning Tree
 - (ii) Fractional Knapsack
 - (iii) Job Sequencing with Deadlines
 - (iv) 0/1 Knapsack
- (f) In the Branch-and-Bound approach for the Traveling Salesman Problem (TSP), bounding is used to
- (i) Find the exact optimal solution immediately
 - (ii) Convert TSP into a dynamic programming problem
 - (iii) Prune paths that cannot yield better solutions
 - (iv) Sort cities by distance
- (g) In the Matrix Chain Multiplication problem, the DP table stores
- (i) The sum of matrix elements
 - (ii) Minimum number of scalar multiplications needed for each subchain
 - (iii) Maximum determinant of matrix chain
 - (iv) Sequence of matrices in ascending order
- (h) In 0/1 Knapsack DP solution, the state $K[i][w]$ represents
- (i) Maximum profit using the first i items for weight limit w
 - (ii) Total number of items used
 - (iii) Minimum weight required to achieve profit i
 - (iv) Fraction of items included

- (i) Topological sorting is applicable to
- (i) Any graph
 - (ii) Directed Acyclic Graph (DAG) only
 - (iii) Undirected graphs only
 - (iv) Cyclic graphs only
- (j) An approximation algorithm guarantees
- (i) Exact optimal solution
 - (ii) Incorrect solution always
 - (iii) Random solution without bounds
 - (iv) Solution within a known factor of optimal

Q.2 (a) Solve the following recurrence relations using the Substitution Method: [7]

- (i) $T(n) = 2T(n/2) + n, T(1)=1$
- (ii) $T(n) = T(n-1) + n, T(1)=1$

(b) State and explain the Master's Theorem. Apply it to solve the following recurrence relation: $T(n)=3T(n/2)+n$ [7]

Q.3 (a) Describe the Merge Sort algorithm with a suitable example. Derive its time complexity using recurrence relations. [7]

(b) Apply the Binary Search algorithm to find the key 41 in the sorted array [5,12,18,25,33,41,56,63,70] Also analyze its time complexity. [7]

Q.4 (a) Apply the Heap Sort algorithm to the array. [16, 14, 10, 8, 7, 9, 3, 2, 4, 1]. Show the initial heap, each deletion step, and the final sorted array. Also analyse the time complexity. [7]

(b) Compare Greedy, Dynamic Programming, and Backtracking approaches with respect to solution strategy, efficiency, and applicability. [7]

Q.5 (a) A thief has a knapsack of capacity 50 kg. The items available are given below. Apply the Fractional Knapsack Algorithm to select items (or fractions) that maximize total value, and compute the total weight and total value in the knapsack. [7]

Item	Weight (kg)	Value (\$)
1	10	60
2	20	100
3	30	120
4	15	75
5	25	80
6	18	90
7	12	48
8	8	32

Explain why this approach guarantees an optimal solution for the fractional knapsack problem.

(b) Apply Bellman Ford's algorithm step-by-step on the given weighted matrix W to compute single-source shortest paths from node 3. State the time complexity of the algorithm. [7]

	1	2	3	4	5
1	0	1	8	1	4
2	1	0	12	4	9
3	8	12	0	7	3
4	1	4	7	0	2
5	4	9	3	2	0

- Q.6 (a) Discuss Dijkstra's algorithm in detail with the help of an example. Explain why the algorithm fails for graphs containing negative edge weights. Also, state the time complexity of Dijkstra's algorithm. [7]
- (b) Discuss the Dynamic Programming approach for solving the Fibonacci series problem. Explain how it improves efficiency over the recursive method. [7]

- Q.7 (a) Given the distance matrix for 5 cities shown below, apply the Dynamic Programming approach to solve the Travelling Salesman Problem. Find the minimum cost tour starting and ending at City 1 and list the optimal tour. Show the key steps involved in computing the solution. [7]

	1	2	3	4	5
1	0	10	15	20	25
2	10	0	35	25	30
3	15	35	0	30	20
4	20	25	30	0	15
5	25	30	20	15	0

- (b) Given the directed graph $G(V,E)$ with vertices $V = \{1, 2, 3, 4, 5\}$ and edges $E = \{(1, 2), (1, 3), (2, 4), (3, 4), (4, 5)\}$: [7]
- (i) Represent the graph using an adjacency matrix.
- (ii) Represent the same graph using an adjacency list.

- Q.8 (a) What is a strongly connected component? Find all the strongly connected components in the following graph. Show all necessary steps and data structure values of the algorithm. [7]



- (b) What is P, NP, NP hard, and NP complete? Write the relationship between all of them. [7]

- Q.9 Write short note on **any two** of the following: [7 x 2 = 14]
- (a) Huffman Coding Algorithm
- (b) Heuristics
- (c) BFS vs DFS
- (d) Cook's theorem



Bihar Engineering University, Patna
End Semester Examination - 2023

Course: B.Tech.
Code: 105402

Semester-IV
Subject: Design and Analysis of Algorithms

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are NINE questions in this paper.
- (iii) Attempt FIVE questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[2 x 7 = 14]**

- (a) The fractional Knapsack problem can be solved by using
 - (i) Greedy method
 - (ii) Divided and conquer method
 - (iii) Dynamic programming
 - (iv) None of these
- (b) BFS on a graph $G = (V, E)$ has running time
 - (i) $O(|V| + |E|)$
 - (ii) $O(|V|)$
 - (iii) $O(|E|)$
 - (iv) None of the above
- (c) The minimum number of colors needed to color a graph having $n > 3$ vertices and 2 edges is
 - (i) 2
 - (ii) 3
 - (iii) 4
 - (iv) 1
- (d) Complexity the recurrence relation $T(n) = 8T\left(\frac{n}{2}\right) + n^2$ is
 - (i) $O(n)$
 - (ii) $O(n^2)$
 - (iii) $O(\log_2 n)$
 - (iv) $O(n^3)$
- (e) Travelling salesman problem belongs to
 - (i) P class
 - (ii) NP class
 - (iii) NP-hard
 - (iv) NP-complete class
- (f) Kruskal's algorithm uses _____ and Prim's algorithm uses _____ in determining the MST
 - (i) Edges, vertex
 - (ii) vertex, edges
 - (iii) Edges, edges
 - (iv) Vertex, vertex
- (g) Level order traversal of a rooted tree can be done by starting from root and performing
 - (i) Depth first search
 - (ii) Breadth first search
 - (iii) Pre-order traversal
 - (iv) In-order traversal
- (h) An algorithm is made up of two independent time complexities $f(n)$ and $g(n)$. Then the complexities of the algorithm is in order of
 - (i) $f(n) \times g(n)$
 - (ii) $\max(f(n), g(n))$
 - (iii) $\min(f(n), g(n))$
 - (iv) $f(n) + g(n)$
- (i) Which of the following standard algorithms is not a greedy algorithm?
 - (i) Dijkstra's shortest path algorithm
 - (ii) Kruskal algorithm
 - (iii) Bellman ford shortest path algorithm
 - (iv) Prim's algorithm
- (j) The node removal of which makes a graph disconnected is called
 - (i) Pendant vertex
 - (ii) Bridge
 - (iii) Articulation point
 - (iv) Coloured vertex



- Q.2 (a) Discuss the average, worst best time complexity of the algorithm. Give suitable examples. [7]
- (b) Write the algorithm for quick-sort and find its complexity. [7]

- Q.3 (a) Construct the Huffman coding tree for the text of characters with given frequencies. [7]
- | | | | | | | | | | | |
|-------------|----|----|----|---|----|----|----|----|----|---|
| Characters | T | I | V | K | L | E | Q | Z | P | R |
| Frequencies | 43 | 38 | 16 | 8 | 56 | 12 | 41 | 13 | 22 | 6 |
- (b) State the general Knapsack problem. Write a greedy algorithm for this problem and derive its time complexity. [7]

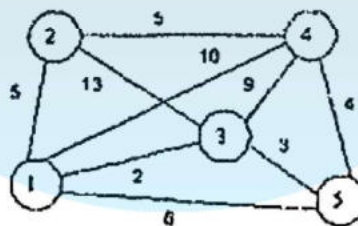
- Q.4 (a) State master's theorem and find the time complexity for the following recurrence: $T(n) = 2T(n^{1/2}) + \log n$ [6]
- (b) What is negative weight-cycle? Write Bellman-Ford algorithm to find single source shortest distance of a directed graph. [8]

- Q.5 (a) Find the minimum number of operation required for the following matrix chain multiplication using dynamic programming $A(10 \times 20) * B(20 \times 50) * C(50 \times 1) * D(1 \times 100)$ [7]
- (b) Write Knuth-Morris-Pratt algorithm for string matching problem. [7]

- Q.6 (a) Write an algorithm to find a minimum spanning tree (MST) for an undirected graph. Estimate the time complexity of your algorithm. [7]
- (b) Using greedy strategy. Schedule the following jobs within deadline so as to maximize the profit. Deadline and profits are mentioned as follow: [7]

Job i	1	2	3	4
Deadline d	3	2	3	1
Profit g	9	7	7	2

- Q.7 (a) Write an algorithm for n-queen's problem find its time-complexity and explain the algorithm using an example. <https://www.akubihar.com> [7]
- (b) Solve the single source shortest path problem for the following graph considering '1' as the source vertex using Dijkstra's algorithm. [7]



- Q.8 (a) Define the classes P and NP. [2]
- (b) Discuss what you mean by polynomial reduction. [2]
- (c) Discuss diagrammatically the relation among P class, NP class, NP hard and NP complete. [2]
- (d) Describe Clique Decision Problem (CDP) [2]
- (e) Explain the max-flow min-cut theorem with an example. [6]

- Q.9 Write short notes on any two of the following: [7x2=14]
- Asymptotic notations
 - Heap creation technique
 - Strassen's matrix multiplication
 - Divide-and-Conquer vs Dynamic programming.



Code : 105402

(2)

B.Tech 4th Semester Exam., 2022

(New Course)

DESIGN AND ANALYSIS OF ALGORITHMS

Time : 3 hours

Full Marks : 70

Instructions :

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

1. Choose the correct answer of the following
(any seven) : $2 \times 7 = 14$

- (a) Any decision trees that sorts n elements has height
 - (i) $\Omega(\lg n)$
 - (ii) $\Omega(n)$
 - (iii) $\Omega(n \lg n)$
 - (iv) $\Omega(n^2)$

(b) Which one of the following is an application of Queue Data Structure?

- (i) When a resource is shared among multiple consumers
- (ii) When data is transferred asynchronously (data not necessarily received at same rate as sent) between two processes
- (iii) Load balancing
- (iv) All of the above

(c) The complexity of binary search algorithm is

- (i) $O(n)$
- (ii) $O(\log n)$
- (iii) $O(n^2)$
- (iv) $O(n \log n)$

(d) In linear search algorithm the worst case occurs when

- (i) the item is somewhere in the middle of the array
- (ii) the item is not in the array at all
- (iii) the item is the last element in the array
- (iv) the item is the last element in the array or is not there at all

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(Turn Over)

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(Continued)



(3)

(e) Given an unsorted array. The array has this property that every element in array is at most k distance from its position in sorted array where k is a positive integer smaller than size of array. Which sorting algorithm can be easily modified for sorting this array and what is the obtainable time complexity?

- (i) Insertion sort with time complexity $O(kn)$
- (ii) Heap sort with time complexity $O(n \log k)$
- (iii) Quick sort with time complexity $O(k \log k)$
- (iv) Merge sort with time complexity $O(k \log k)$

(f) Approach of dynamic programming is similar to

- (i) parsing
- (ii) hash table
- (iii) divide and conquer algorithm
- (iv) greedy algorithm

(4)

(g) In the divide and conquer process, breaking the problem into smaller sub-problems is the responsibility of

- (i) divide/break
- (ii) sorting/divide
- (iii) conquer/solve
- (iv) merge/combine

(h) A priority queue is implemented as a Max-heap. Initially it has 5 elements. The level order traversal of the heap is 10, 8, 5, 3, 2. Two new elements '1' and '7' are inserted into the heap in that order. The level order traversal of the heap after the insertion of the elements is

- (i) 10, 8, 7, 5, 3, 2, 1
- (ii) 10, 8, 7, 2, 3, 1, 5
- (iii) 10, 8, 7, 1, 2, 3, 5
- (iv) 10, 8, 7, 3, 2, 1, 5

(i) If a problem can be solved by combining optimal solutions to non-overlapping problems, the strategy is called

- (i) dynamic programming
- (ii) greedy
- (iii) divide and conquer
- (iv) recursion



(5)

- (j) The choice of polynomial class has led to the development of an extensive theory called
- (i) computational complexity
 - (ii) time complexity
 - (iii) problem complexity
 - (iv) decision complexity
2. (a) Calculate the time complexity of the following problem using divide and conquer strategies : 7
- (i) $T(n) = \sqrt{n}T(\sqrt{n}) + n, n > 2$
 $= a, n \leq 2$
 - (ii) $T(n) = T(n-1) + 1/n, n > 1$
 $= 1, n = 1$
- (b) Write time function and calculate the time complexity, space complexity and number of function calls of the following pseudocode using substitution method : 7

```

rec (n) {
    if (n ≤ 1)
        return(1);
    else {
        rec (n/2);
        for (i = 1; i ≤ n; i++)
            printf("algorithm");
    }
}

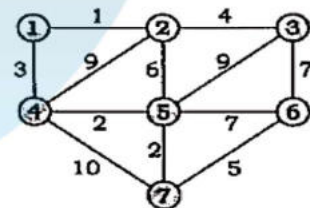
```

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(Turn Over)

(6)

3. (a) Explain the working of merge sort algorithm with an example. Give the complexity calculation of merge sort. 7
- (b) What are the rules of manipulate Big-Oh expression and about the typical growth rates of algorithms. 7
4. (a) What is heap sort? What is the effect of calling MAX-HEAPIFY(A, i) when the element A[i] is larger than its children? 7
- (b) Explain how radix sort works, to what inputs it can be applied and what is its asymptotic complexity? 7
5. What do you mean by optimal solution in greedy approach? Define the properties and function of greedy approach. Consider the graph $G = (V, E)$ given below. Find the minimum spanning tree by Prim's algorithms. 14



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(Continued)



(7)

6. Find the optimal way to multiply the following matrices to perform the fewest multiplications :

14

Matrix	A1	A2	A3	A4
Dimension	5×11	11×4	4×15	15×23

7. You are given a graph containing n vertices and m edges and given that the graph doesn't contain cycle of odd length. What is the time complexity of the best known algorithm to find out whether the graph is bipartite or not?

14

8. What is activity selection problem? Suppose that instead of always selecting the first activity to finish, we select the last activity to start that is compatible with all previously selected activities. Describe how this approach is a greedy algorithm, and prove that it yields an optimal solution.

14

9. Write short notes on the following :

14

(a) Cook's theorem

(b) Approximation algorithms

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